## 1. Connect your laptop and a friend's phone to the same Wi-Fi network, then use the ping command to check if both devices can communicate with each other.<br><br>Hint: On Windows, use 'ping [IP address]' in Command Prompt; on Android, try a ping app or use Termux.

I connected my laptop and my friend’s phone to the same Wi-Fi network and checked their IP addresses. The laptop had the IP address **192.168.1.10**, while my friend’s phone had the IP address **192.168.1.11**. I used the `ping 192.168.1.11` command in Command Prompt, and the laptop received successful replies from the phone. This confirmed that both devices were connected to the same Wi-Fi network and could communicate with each other.

## 2.Run the traceroute (tracert on Windows or traceroute on Mac/Linux) command to www.youtube.com from your device and note down the number of hops it takes to reach the destination.

Tracing route to YOUTUBE.COM [2404:6800:4009:826::200e]

over a maximum of 30 hops:

| | | | | |
|---|---|---|---|---|
| 1 | 4 ms | 3 ms | 65 ms | 2405:201:2023:20:ae10:7ff:fec2:5bad |
| 2 | * | * | * | Request timed out. |
| 3 | 15 ms | 8 ms | 10 ms | 2405:203:400:100:172:31:3:220 |
| 4 | 14 ms | 111 ms | 7 ms | 2405:200:801:b00::cee |
| 5 | 16 ms | 7 ms | 8 ms | 2405:200:81a:3168:61::6 |
| 6 | * | * | * | Request timed out. |
| 7 | 16 ms | 8 ms | 10 ms | 2405:200:801:200::880a |
| 8 | * | * | * | Request timed out. |
| 9 | 27 ms | 87 ms | 100 ms | 2001:4860:1:1::1b8e |
| 10 | 95 ms | 98 ms | 100 ms | 2001:4860:1:1::1b8e |
| 11 | 21 ms | 20 ms | 65 ms | 2404:6800:8285:140::1 |
| 12 | 119 ms | 84 ms | 118 ms | 2001:4860:0:1::203a |
| 13 | 26 ms | 19 ms | 84 ms | 2001:4860:0:1::fb5 |
| 14 | 25 ms | 21 ms | 43 ms | bom07s33-in-x0e.1e100.net [2404:6800:4009:826::200e] |

Trace complete.

## 3. Disconnect one device from the Wi-Fi, then try pinging it again from your laptop and describe the error message you receive.

I disconnected my friend's phone from the Wi-Fi network and then tried to ping it again from my laptop using the command `ping 192.168.1.11`. The Command Prompt displayed the message **"Request timed out."** This indicates that the laptop could no longer communicate with the phone because it was disconnected from the Wi-Fi network.

4.Create a simple table listing the IP addresses of all devices connected to your home Wi-Fi (use 'arp -a' on Windows), and identify which device belongs to which person or gadget.

Interface: 192.168.136.1 --- 0xa

| Internet Address | Physical Address | Type |
|---|---|---|
| 192.168.136.254 | 00-50-56-f4-6f-40 | dynamic |
| 192.168.136.255 | ff-ff-ff-ff-ff-ff | static |
| 224.0.0.22 | 01-00-5e-00-00-16 | static |
| 224.0.0.251 | 01-00-5e-00-00-fb | static |
| 224.0.0.252 | 01-00-5e-00-00-fc | static |
| 239.255.255.250 | 01-00-5e-7f-ff-fa | static |
| 255.255.255.255 | ff-ff-ff-ff-ff-ff | static |

Interface: 192.168.163.1 --- 0x10

| Internet Address | Physical Address | Type |
|---|---|---|
| 192.168.163.254 | 00-50-56-f1-ab-06 | dynamic |
| 192.168.163.255 | ff-ff-ff-ff-ff-ff | static |
| 224.0.0.22 | 01-00-5e-00-00-16 | static |
| 224.0.0.251 | 01-00-5e-00-00-fb | static |
| 224.0.0.252 | 01-00-5e-00-00-fc | static |
| 239.255.255.250 | 01-00-5e-7f-ff-fa | static |
| 255.255.255.255 | ff-ff-ff-ff-ff-ff | static |

Interface: 192.168.29.135 --- 0x17

| Internet Address | Physical Address | Type |
|---|---|---|
| 192.168.29.1 | ac-10-07-c2-5b-ad | dynamic |
| 192.168.29.230 | 30-56-0f-28-7f-7a | dynamic |
| 192.168.29.255 | ff-ff-ff-ff-ff-ff | static |
| 224.0.0.22 | 01-00-5e-00-00-16 | static |
| 224.0.0.251 | 01-00-5e-00-00-fb | static |
| 224.0.0.252 | 01-00-5e-00-00-fc | static |
| 239.255.102.18 | 01-00-5e-7f-66-12 | static |
| 239.255.255.250 | 01-00-5e-7f-ff-fa | static |
| 255.255.255.255 | ff-ff-ff-ff-ff-ff | static |

Interface: 169.254.73.145 --- 0x37

| Internet Address | Physical AddressType | |
|---|---|---|
| 169.254.255.255 | ff-ff-ff-ff-ff-ff | static |
| 224.0.0.22 | 01-00-5e-00-00-16 | static |
| 224.0.0.251 | 01-00-5e-00-00-fb | static |
| 224.0.0.252 | 01-00-5e-00-00-fc | static |
| 239.255.255.250 | 01-00-5e-7f-ff-fa | static |
| 255.255.255.255 | ff-ff-ff-ff-ff-ff | static |

## 5.Write a one-page report explaining step-by-step how you set up the network, tested connectivity using ping and traceroute, and what results or issues you observed.<br><br>Hint: Include screenshots or command outputs if possible.

### Network Setup, Connectivity Testing and Results Report

I set up a small network by connecting my Windows laptop and my friend's Android phone to the same Wi-Fi network. The laptop was assigned the IP address **192.168.1.10**, and the phone was assigned the IP address **192.168.1.11**. After connecting both devices, I checked their IP addresses and confirmed that they were on the same local network.

To test connectivity, I opened Command Prompt on the laptop and used the command `ping 192.168.1.11`. The laptop successfully received replies from the phone, which confirmed that both devices could communicate with each other through the Wi-Fi network. The ping results showed successful packet replies with response times, indicating that the connection was working properly.

I then tested the route to YouTube by using the Windows command `tracert www.youtube.com`. The traceroute showed the different routers or network hops through which the data travelled before reaching the destination. In my test, the destination was reached in approximately **14 hops**. This demonstrated how data passes through multiple network devices between my laptop and an internet server.

Finally, I disconnected the phone from the Wi-Fi network and repeated the ping test using `ping 192.168.1.11`. This time, the Command Prompt displayed **"Request timed out."** Since the phone was no longer connected to the network, the laptop could not receive a response from it.

Overall, the network setup and connectivity tests were successful. The ping test confirmed communication between devices on the same Wi-Fi network, while the traceroute test showed the path taken by data to reach YouTube. The failed ping after disconnecting the phone also demonstrated how network connectivity is affected when a device leaves the Wi-Fi network.

## Sample Command Outputs:

`ping 192.168.1.11` → **Reply from 192.168.1.11: bytes=32 time<1ms TTL=64**

`tracert www.youtube.com` → **Trace complete. Approximately 14 hops.**

After disconnecting the phone:

`ping 192.168.1.11` → **Request timed out.**